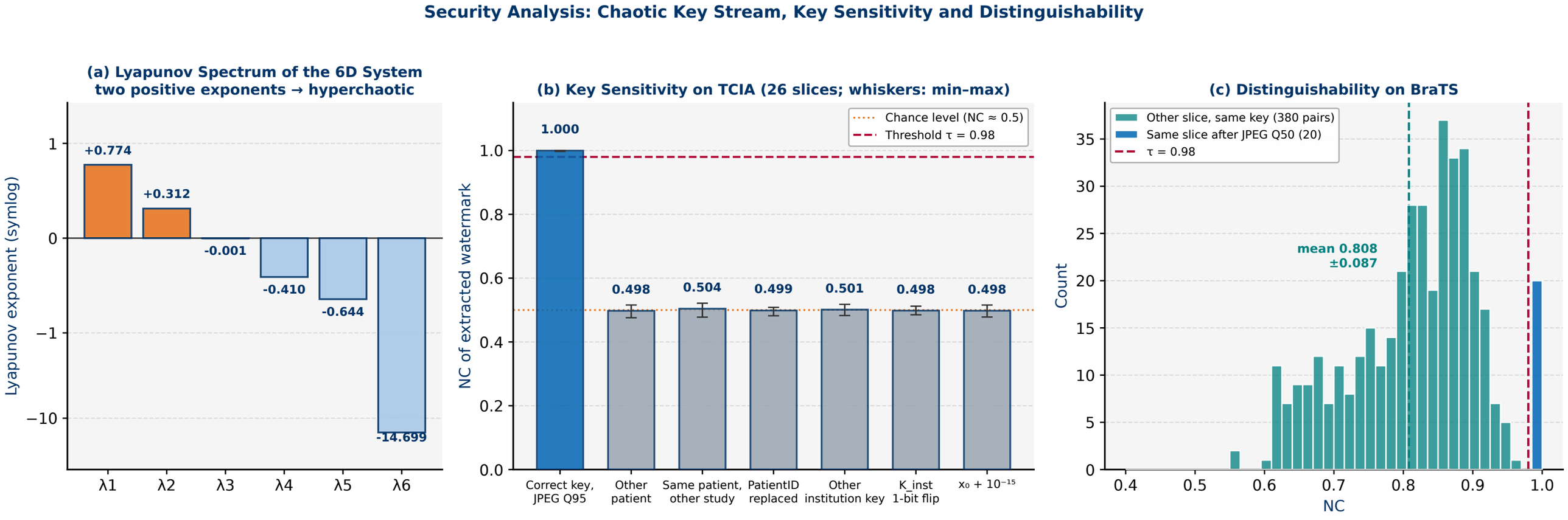
Security Analysis: Chaotic Key Stream, Key Sensitivity and Distinguishability
(a) Lyapunov Spectrum of the 6D System
two positive exponents → hyperchaotic
Lyapunov exponent (symlog)
+0.774
+0.312
-0.001
-0.410
-0.644
-14.699
λ1
λ2
λ3
λ4
λ5
λ6
(b) Key Sensitivity on TCIA (26 slices; whiskers: min–max)
NC of extracted watermark
Chance level (NC ≈ 0.5)
Threshold τ = 0.98
1.000
0.498
0.504
0.499
0.501
0.498
0.498
Correct key, JPEG Q95
Other patient
Same patient, other study
PatientID replaced
Other institution key
K_inst 1-bit flip
x₀ + 10⁻¹⁵
(c) Distinguishability on BraTS
Other slice, same key (380 pairs)
Same slice after JPEG Q50 (20)
τ = 0.98
mean 0.808
±0.087
Count
NC